LabTech Testing Limited

STANDARD TEST PROCEDURE

Plate Load Test

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IMPLEMENTATION DATE _____

PREPARED BY : _____

APPROVED BY : _____

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1 Scope

This procedure covers equipment setup, incremental loading, settlement measurement, reporting and associated calculations for in-situ plate load tests. The method basis is BS 1377: Part 9: 1990, Clause 4.1, supplemented by the applicable project specification. For relevant work, apply the Buildings Department Code of Practice for Foundations 2017 (2024 Edition), Clause 8.2, read with applicable amendments, and Housing Authority General Specification for Building Works 2024, EXC.T910.1. Equipment control shall follow HOKLAS SC-02 and SC-16. Confirm the method edition, contract specification, approved test arrangement, loading programme and reporting requirements before testing. The settlement and modulus relationships in Section 7 apply to cohesionless soils as described in the cited Code.

2 Equipment

2.1 Load Measuring System

The loading and measurement system shall have sufficient capacity for the maximum specified test load. Confirm the reaction system against the approved setup before testing. Where HA GS EXC.T910.1 applies, use the kentledge arrangement approved by the Contract Manager (CM). The jack and load cell shall remain vertical and concentric with the test plate throughout loading.

- Hydraulic Jack and Pump: A hydraulic loading system of adequate capacity to apply the required test loads.

- Load Cell: Use a calibrated load cell and compatible indicator with an accuracy within 1% of the applied load over the working range. In accordance with HOKLAS SC-16, calibrate the force-measuring device before each test or series of tests on the same site through a competent calibration body as defined in HOKLAS SC-02. Carry out post-test calibration where required by the contract. Record instrument identification, calibration status, applicable corrections and the zero/tare check before use.

- Load Indicator / Data Logger: A compatible digital indicator or data logger to display the load readings.

2.2 Deformation Measurement System

- Dial Gauges: Use four gauges at orthogonal positions, with sufficient travel for the anticipated plate movement. Secure the gauges to the test plate, with their tips bearing vertically on the independent reference beam. Resolution shall be 0.01 mm per division or better; where HA GS EXC.T910.1 applies, accuracy shall be 0.01 mm. Under HOKLAS SC-16, displacement devices

shall be calibrated at intervals not exceeding one year and checked at one point against an appropriate reference before use.

- Mounting Accessories: Suitable magnetic bases, clamps and fixtures to secure the dial gauges to the test plate.

2.3 Loading Plate

A rigid square or circular steel plate, not less than 300 mm in side length or diameter and at least 25 mm thick, shall be used. The plate shall have adequate rigidity for the maximum test load. The underside of the plate shall be nominally flat. The plate should have a centre mark or guide to assist in aligning the hydraulic jack, helping to distribute the load evenly and minimize tilting.

2.4 Timer

A digital timer, stopwatch, or an integrated time-recording function within the data logger to accurately record the elapsed time during load increments and monitor the rate of settlement.

2.5 Thermometer (Conditional)

Provide a thermometer for temperature records where required by the contract, including HA GS EXC.T910.1. Maintain its calibration in accordance with the applicable equipment-control requirements; HOKLAS SC-16 specifies a maximum two-year interval for temperature-measuring devices. Protect the reference beam from direct sun and significant temperature effects.

3 Preparation

3.1 Site Condition Verification

Testing shall be undertaken by authorised personnel under the laboratory's controlled competence and supervision arrangements. Before setup, verify the specified test location, coordinates, level, soil description and groundwater conditions, and visually check the prepared test area:

- The test area for the plate should be generally level and as undisturbed as possible.
- All loose material and embedded fragments must be carefully trimmed off and removed.
- The test area should be adequately protected from adverse weather, excessive moisture changes, or direct sunlight.

3.2 Plate Bedding

To ensure uniform load distribution, the test surface shall be properly levelled. A thin layer of suitable bedding material (such as clean dry sand or quick-setting plaster, depending on the site soil condition) shall be applied to fill any hollows and ensure the loading plate is in full and uniform contact with the test surface.

3.3 Equipment Erection

- Place the loading plate centrally onto the prepared bedding layer.

- Position the loading column (hydraulic jack and load cell) centrally over the plate and ensure it is vertical so that the reaction load is applied directly to the plate without eccentricity.

- Secure the four dial gauges to the plate at orthogonal positions, with their measuring tips on firm reference-beam targets. Record initial readings, the reading direction and the settlement datum before loading. Use a consistent datum throughout the cycles and record any instrument adjustment.

- Crucial Step**:** Take care not to preload the test plate during the erection of the reaction loading and force measurement systems.

- Support the independent reference beam on two rigid foundations outside the influence zones of both the loaded plate and the kentledge reaction area. Confirm the support separations against the adopted method and approved layout, record the actual positions, and check beam stability before readings. Complete the deformation-measurement setup as the last erection operation.

- Where necessary, rigid steel packing plates or spacers may be used to close the gap between the loading assembly and the reaction beam. They shall be centrally aligned, adequately supported and restrained against slipping or tilting, and capable of sustaining the maximum test load without undue deformation or buckling. The connection between the loading column and the test plate shall resist plate tilting; no ball seating shall be provided at this connection. No additional preload shall be imposed during erection.

- Record the packing arrangement, dimensions and mass, the mass supported by the plate, and the load-cell zero/tare procedure. Add self-weight omitted from the force reading without double counting. Record both indicated load and corrected plate pressure, including the actual initial self-weight pressure at nominal zero jack load.

- The actual plate shape, dimensions and contact area shall be recorded. Applied pressure shall be calculated from the corrected force divided by the actual plate contact area.

4 Test Procedure

4.1 Initial Seating Load

Before formal loading, an initial seating pressure not exceeding 25% of the initial overburden pressure may be briefly applied to reduce bedding error. Record the seating load, corresponding pressure, duration and initial gauge readings.

4.2 Standard Loading Sequence

Confirm the working test load P0 (kN) and plate area A (m^2) before testing. In this procedure, W denotes the corresponding test pressure, W = P0/A (kPa). Use increments of 0.5W in three cycles to W, 2W and 3W; corresponding loads are P0, 2P0 and 3P0. Under HA GS EXC.T910.1, P0 is the allowable working pressure multiplied by plate area (the load denoted W in that specification). When applying BD Clause 8.2(2), confirm P0 against the plate test load Wt defined in Clause 8.2(2)(b). Record the adopted load basis and approved programme; apply any specified higher maximum load.

Cycle 1: Apply pressure in increments of 0.5W up to 1.0W, then release in decrements of 0.5W back to 0W.

Cycle 2: Reapply pressure in increments of 0.5W up to 2.0W, then release in decrements of 0.5W back to 0W.

Cycle 3: Reapply pressure in increments of 0.5W up to 3.0W, maintain the maximum stage as specified, then unload in the recorded 0.5W steps.

Cycle	Pressure	Loading Sequence
1	1.0W	0W ~ 0.5W ~ 1.0W ~ 0.5W ~ 0W
2	2.0W	0W ~ 0.5W ~ 1.0W ~ 1.5W ~ 2.0W ~ 1.5W ~ 1.0W ~ 0.5W ~ 0W
3	3.0W	0W ~ 0.5W ~ 1.0W ~ 1.5W ~ 2.0W ~ 2.5W ~ 3.0W ~ 2.5W ~ 2.0W ~ 1.5W 1.0W ~ 0.5W ~ 0W

4.3 Maintenance of Load and Settlement Records

- For each load increment, the load shall be kept constant.

- Settlement Readings: Record the time, indicated load and all four gauge readings at the start of the test and at load changes. During each stage, take readings at the approved intervals, normally every 5 minutes, with more frequent observations at the start of each increment.

- Increment Holding Time: Maintain each load stage for at least 10 minutes and until the settlement rate is less than 0.05 mm in 10 minutes. Keep the load constant and retain the readings used to confirm the rate.

- Maximum Load Holding Time: Maintain the maximum test stage for at least 72 hours under BD Clause 8.2(2)(d) and HA GS EXC.T910.1, or longer if specified. Record readings at the start, end and approved intermediate intervals. Confirm the observation schedule before loading; 10:00, 13:00 and 17:00 readings are an example only.

4.4 Environmental Records

Record temperature and weather during the test where required by the contract, including HA GS EXC.T910.1, and during prolonged holding periods as specified. Record relevant changes in groundwater, moisture, rainfall or shading that may affect the readings.

5 Reporting Results

Prepare a traceable test report identifying the project, client, test location, test dates, adopted method and edition, and the results. Include the specified calculations and interpretation, supporting records and technical review. The report shall contain the following information and any additional contract requirements:

5.1 Graphical Presentations:

- Plot the graph of applied pressure and settlement versus time elapsed.
- Plot the graph of settlement versus applied pressure.

5.2 Detailed Tabular Records:

- Record stage identification, applied load and pressure, actual times and elapsed duration, including maximum-load holding and unloading.
- Record the four individual gauge readings, reading direction, calculated settlements, mean settlement and rate checks. Include the plate shape, dimensions and area, self-weight correction and zero/tare basis. Identify interruptions, adjustments or deviations and their treatment in the results.

5.3 Equipment and Calibration Validation:

- Identify the load-measuring and displacement devices, their calibration dates and certificates, the pre-use check results, and applicable calibration/correlation charts. Retain the supporting records and include calibration details required by the contract.

5.4 Site Documentation:

- Include the test coordinates and reduced level, description of the tested soil stratum, test dates, temperature and weather. Describe the reaction system and include setup drawings and photographs of the surface, plate, load train, reference beam and supports. Where HA GS EXC.T910.1 applies, include soil-modulus results and interpretation in accordance with the agreed reporting requirements.

6 Security of Testing Equipment

Maintain a stable, protected test environment throughout loading and the maximum-load holding period.

- Environmental Control**:** The test area shall be free of water accumulation.
- Equipment Protection: Security to the testing equipment should be ensured at all times during the test.
- *Confirm arrangements for load maintenance, observations, site security and weather protection before testing. Maintain these arrangements throughout the holding period. Record any disturbance, load interruption or movement of the reference system, and obtain technical review before accepting affected readings or continuing the test.*

7 Reporting and Reference Calculations

Report the measured loading and settlement results together with the calculations and interpretation required by the applicable project specification. Record the adopted parameters, units, load level and supporting references.

7.1 Reference Settlement Limit:

- For cohesionless soils, Clause 8.2(2)(f) of the Code of Practice for Foundations 2017 (2024 Edition) gives the maximum settlement criterion Smax ≤ 0.15B. Smax denotes the maximum settlement during the test, not the residual settlement after unloading.
- For a 300 mm test plate, 0.15B corresponds to 45.0 mm.

7.2 Reference Soil Modulus Equations:

Clause 8.2(2)(g) gives the following relationships for obtaining the soil Young's modulus from the measured settlement at test load Wt. These relationships are provided for engineering reference for cohesionless soils.

7.2.1 Plate Shape:

- Circular test plate: $E_s = \frac{W_t(1-\nu^2)}{S \times B}$
- Square test plate: $E_s = \frac{W_t(1-\nu^2)}{1.13 \times S \times B}$

Rearranged Relationship for a Circular Plate:

- For a specified soil modulus, rearrange the circular-plate relationship as shown below to calculate the corresponding settlement at test load Wt.
- Use Wt in kN, B in m and Es in kPa; the calculated S is in m. Convert measured settlement from mm to m before substitution, and multiply calculated settlement in m by 1000 to express it in mm.

$$S = W_t(1 - \nu^2)/(B \times E_s)$$

Symbols and units used in the reference equations:

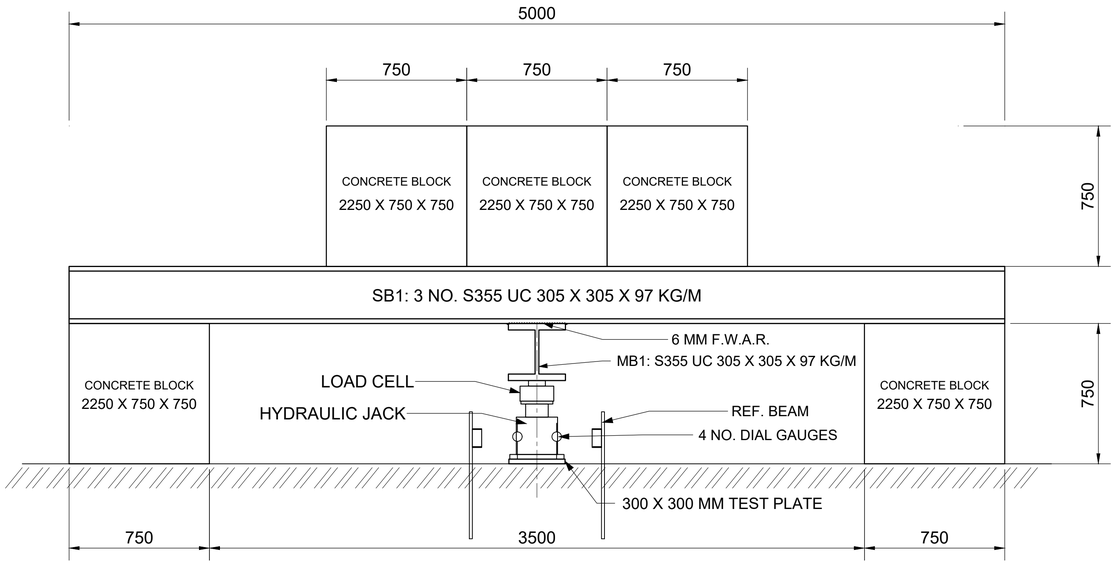
S	=	Settlement of the test plate measured at test load Wt, in m (measured mm divided by 1000).
W_t	=	Load at which settlement S is used in the modulus calculation, in kN; confirm the adopted test-load basis against BD Clause 8.2(2)(b) and the approved project programme.
B	=	Width of a square plate or diameter of a circular plate, in m.
Es	=	Young's modulus of the soil, in kPa.
ν	=	Poisson's ratio of the tested soil, dimensionless; record the value adopted for the calculation and its basis.

Record the selected Poisson's ratio and its basis. Where BD Clause 8.2(2)(h) applies, document comparison of the calculated soil modulus with the design value. Technically review calculations, interpretation and any specified criteria against the applicable project requirements before report issue.

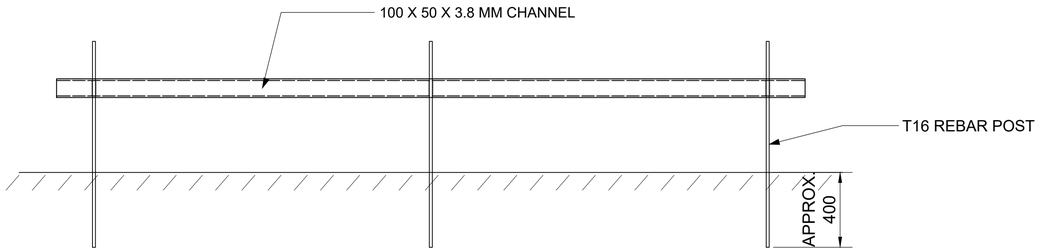
Appendix A

Typical Setup and Design Calculation of Plate Load Test

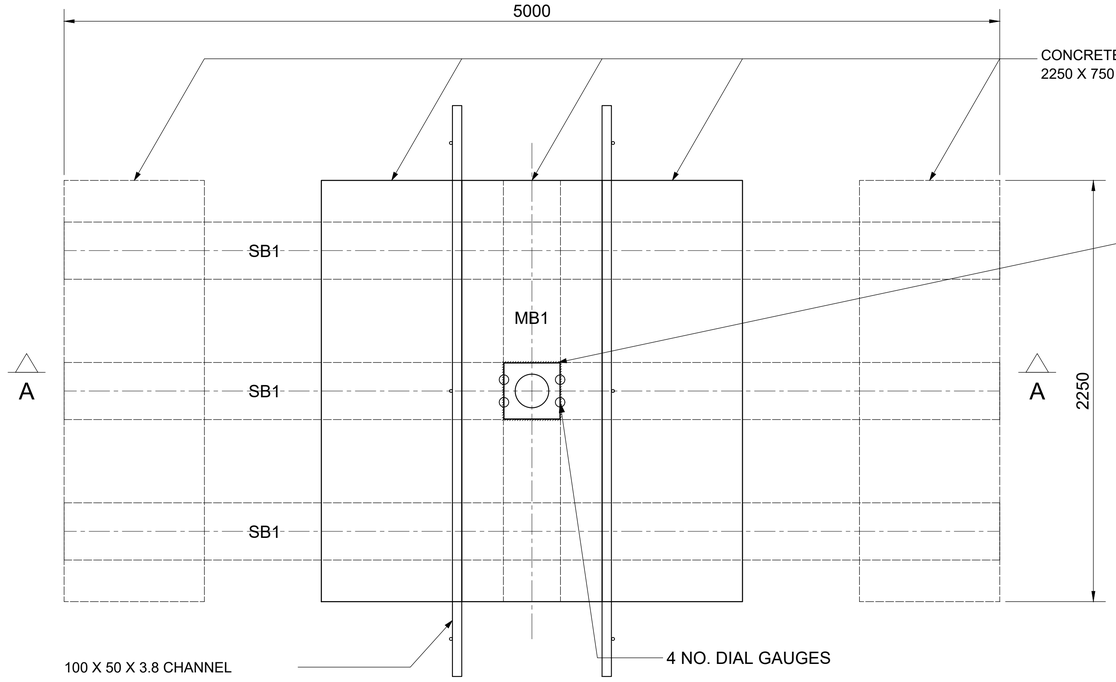
Use the approved project setup and design details for the actual plate, reaction system, packing arrangement and reference-beam supports. Confirm the layout and capacity for the specified maximum test load before erection.



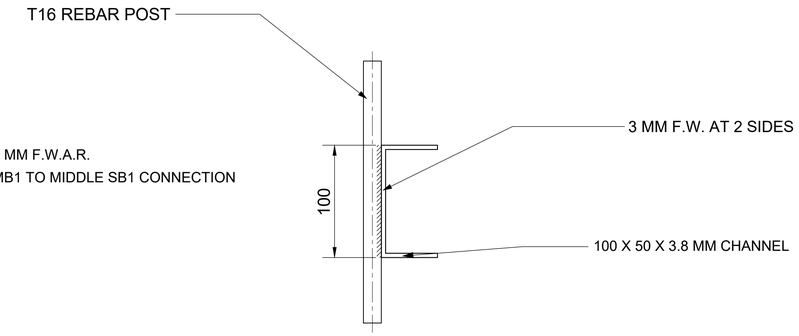
SECTION A-A



REFERENCE BEAM - ELEVATION



PLAN VIEW



CHANNEL / T16 CONNECTION

TYPICAL PLATE LOAD TEST SETUP

NOTES:
1. ALL DIMENSIONS IN MM. ALL CONCRETE BLOCKS: 2250 X 750 X 750 MM.
2. SB1: 3 NO. S355 UC 305 X 305 X 97 KG/M, 5000 LONG.
3. MB1: 1 NO. S355 UC 305 X 305 X 97 KG/M, 2250 LONG.
5. THREE UPPER BLOCKS WITH 2250 DIRECTION TRANSVERSE TO SB1.
6. MB1 TO MIDDLE SB1: 6 MM F.W.A.R. SECURE MB1 BEFORE JACKING.
9. REFERENCE SUPPORTS TO BE INDEPENDENT OF THE REACTION SYSTEM.

LaBTECH			Title:	**Checking of Plate Load Test Setup (Max. 750kPa)**
Date :	Oct-2026	Initial : Checked :	Revision :	v20261021	Page :

Design Assumption

Unit weight of Concrete Block	=	24	kN/m^3
(CoPFD-Cl.2.2.2(5)) Presumed allowable bearing pressure on ground	=	100	kPa

Setup Information

Test pressure at 3W	=	750	kPa
Plate width	=	0.3	m (rectangular)
Plate area	=	0.09	m^2
Test load	=	67.5	kN
Size of concrete block	=	750 x 750 x 2250	
Self-weight of each concrete block	=	30.38	kN
Size of beam SB1	=	UC 305x305x97kg/m	
Length of SB1	=	5.00	m
Self-weight of each SB1	=	4.76	kN
Size of beam MB1	=	UC 305x305x97kg/m	
Length of MB1	=	2.25	m
Self-weight of each MB1	=	2.14	kN

Check: Bearing Pressure of Supporting Concrete Block

Loading

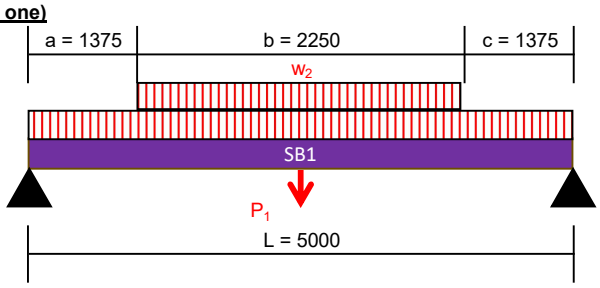
3 nos of concrete block on top	=	91.125	kN
3 nos of SB1	=	14.274	kN
1 nos of MB1	=	2.141	kN
2 nos of supporting concrete block	=	60.750	kN
Total Dead Load (DL)	=	168.290	kN

Base area of each supporting concrete block, A = 0.75 x 2.25 = 1.6875 m^2

Bearing pressure of supporting concrete block = 168.29 / 2nos / 1.69 = 49.9 kPa < 100 kPa OK

LaBTECH		Title:	**Checking of Plate Load Test Setup (Max. 750kPa)**
Date : Oct-2026	Initial : Checked :	Revision : v20261021	Page :

Check: SB1 (middle one)



UC 305x305x97kg/m
S355

Loading

P_1 = Self-weight of MB1
= 1.4 x 2.14
= 3.00 kN (factored)

w_1 = Self-weight of SB1
= 1.4 x 97 x 9.81/1000
= 1.33 kN/m (factored)

w_2 = Self-weight of concrete block on top
= 1.4 x 24 x 0.75 x 1.125
= 28.35 kN/m (factored)

Design Shear, V = $P_1/2 + w_1L/2 + w_2b/L\ (b/2 + c)$
= 3 / 2 + 1.33 x 5 / 2 + 28.35 x 2.25 / 5 x (2.25/2 + 1.375)
= 36.72 kN

Design Moment, M = $P_1L/4 + w_1L^2/8 + w_2\ [(L/2)^2 - a^2]/2$
= 3 x 5 / 4 + 1.33 x 5^2 / 8 + 28.35 x [(5/2)^2 - 1.375^2]/2
= 69.70 kNm

Structural Checking

(STRUCTURAL CHECKING REFER NEXT PAGES)

Welding Checking

Weld Length, Lw = 305.3 x 4 nos
= 1221.2 mm

Weld Size, S = 6 mm F.W.

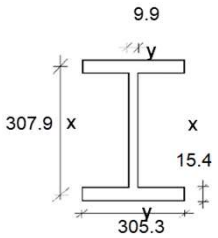
Weld Strength, pw = 220 MPa

Weld Capacity = 0.707 x S x pw x Lw /1000
= 0.707 x 6 x 220 x 1221 /1000
= 1140 kN
> 3.00 kN OK

Calculation Sheet	Job No.	Sheet No.	Rev.
			P.
	Member / Location:	SB1	
Job Tilte :	Drg. Ref.:		
	Made By:	Date: Oct-26	Check By:

DESIGN CODE

Code of Practice for the Structural Use of Steel 2011



MEMBER SPECIFICATION

Member

Steel grade	=	BS S355	
Section	=	UC 305x305x97	(H-section)
Type	=	hot-rolled steel section	

MATERIAL/SECTIONAL PROPERTIES

Modulus of elasticity	E	=	205000	MPa								
Mass	m	=	96.9	kg/m								
Area	A	=	123	cm^2								
Dimensions	D	=	307.9	mm	d	=	246.7	mm				
	B	=	305.3	mm	b	=	152.65	mm				
	T	=	15.4	mm	t	=	9.9	mm				
Moment of inertia	Ix	=	22250	cm^4	Iy	=	7308	cm^4	Iu	= N.A.	cm^4	Iv = N.A. cm^4
Radius of gyration	rx	=	13.4	cm	ry	=	7.69	cm	ru	= N.A.	cm	rv = N.A. cm
Elastic modulus	Zx	=	1445	cm^3	Zy	=	479	cm^3				
Plastic modulus	Sx	=	1592	cm^3	Sy	=	726	cm^3				
	Svx	=	150.6	cm^3	Svy	=	717.7					
Torsional modulus	J	=	91.2	cm^4								
Design strength	py	=	355	MPa								

SECTION CLASSIFICATION

Parameter	ε	=	SQRT(275 / py)		=	0.8801		
Stress ratio	r1	=	Nc / (d * t * py)		=	0		
	r2	=	Nc / (A * py)		=	0		
Dimension ratio	b/T	=	9.9	<= 15 ε	=	13.2	Flange:	Class 3
	d/t	=	24.9	<= 80 ε / (1+r1)	=	70.4	Web:	Class 1
							Overall:	Class 3

DESIGN FORCES AND MOMENTS

Nc (kN)	Nt (kN)	Vx (kN) [along]	Vy (kN) [along]	Mx (kNm) [abt]	My (kNm) [abt]	Tr (kNm)
0.00	0.00	0.00	36.72	69.70	0.00	0.00

SHEAR CAPACITY CHECK

Shear area	Avy	= t * D	=	3048	mm^2					*(Cl. 8.2.1)*	
Shear capacity	Vcy	= py * Avy / SQRT(3)	=	624.76	kN	>=	Vy =	36.72	kN	OK	6%
Shear load check	Vy	<= 0.6 * Vcy	=	374.86	kN		(Low shear case)				
Shear area	Avx	= 0.9 * (2 * B * T)	=	8463	mm^2						
Shear capacity	Vcx	= py * Avx / SQRT(3)	=	1734.6	kN	>=	Vx =	0	kN	OK	0%
Shear load check	Vx	<= 0.6 * Vcx	=	1040.7	kN		(Low shear case)				

MOMENT CAPACITY CHECK

(Cl. 8.2.2)

(Low Shear Case)

Moment capacity	Mcx	= py * Zx	=	512.98	kNm	>=	Mx =	69.7	kNm	OK	14%
Moment capacity	Mcy	= py * Zy	=	170.05	kNm	>=	My =	0	kNm	OK	0%

BIAXIAL MOMENTS: LOCAL CAPACITY CHECK

$$\frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{69.7}{512.98} + \frac{0}{170.045} = 0.14 \quad <= \quad 1$$

OK 14%

Calculation Sheet	Job No.	Sheet No.	Rev.
			P.
	Member / Location:	SB1	
Job Tilte :	Drg. Ref.:		
	Made By:	Date: Oct-26	Check By:

TENSION CAPACITY CHECK

Effective Area $a_e = Ke * a_n$ = 1.1 x(12300 (Ag) - 0 (Area deduction for hole in mm^2)) *(Cl. 8.6.1)*

= 13530 mm^2 >= 12300 mm^2 (Therefore take Ag)

Connection Type: **Welded Connection**

$a_2 = Ag - a_1$ = 9251.8 mm^2

Tension capacity $Pt = py * (a_e - 0 \; a_2)$ = 4366.5 kN >= Nt = 0 kN **OK** 0%

COMPRESSION CAPACITY CHECK *(Cl. 8.7.5)*

Effective length	Lcx =	**5.00** m	Lcy =	**5.00** m	
Slenderness	λx = Lcx / rx		λy = Lcy / ry		
	= 37.3134		= 65.0195		
From Table 8.7, use buckling curve	[b]		[c]		
Robertson constant	α = 3.5		α = 5.5		
	$pEx = (\pi^2 * E) / \lambda x^2$		$pEy = (\pi^2 * E) / \lambda y^2$		
	= 1453		= 479		
Limiting slenderness	λo = 0.2 * SQRT(π^2 * E / py)				
	= 15.0988				
Perry factor	η = α * (λx - λ0) / 1000		η = α * (λy - λ0) / 1000		
	= 0.078		= 0.275		
	ϕc = [py + (η + 1) * pEx] / 2		ϕc = [py + (η + 1) * pEy] / 2		
	= 961		= 482		
Compressive buckling strength	pcx = pEx * py / [ϕc + SQRT(ϕc^2 - pEx * py)]		pcy = pEy * py / [ϕc + SQRT(ϕc^2 - pEy * py)]		
	= 322.742 MPa		= 231.691 MPa		
Compression resistance	Pc = A * pcy = 2849.8 kN	>= Nc = 0 kN		**OK**	0%

LATERAL-TORSIONAL BUCKLING CHECK *(Cl. 8.3.1)*

Effective length	L_{LT} =	**5.00** m	m_{LT} =	**1**	
Slenderness	λ = L_{LT} / ry		= 65.0195		
Dist. betw. flange shear centers	hs = D - T		= 292.5 mm		
Web depth	hw = D - 2 * T		= 277.1 mm		
wraping constant for channel	H =		= 1.9E+12 mm^6		
Torsional index	x = 0.566 * hs * SQRT(A / J)		= 19.2264		
	γ = 1 - Iy / Ix		= 0.67155		
Buckling parameter	u = $(4 * Sx^2 * \gamma)/(A^2 * hs^2)]^{0.25}$		= 0.85161		
Slenderness factor	v = $1 / [1+ 0.05 * (\lambda / x)^2]^{0.25}$		= 0.8931		
	βw = Zx / Sx		= 0.90766		
Equivalent slenderness	λ_{LT} = u * v * λ * SQRT(βw)		= 47.1135		
	pE = $(\pi^2 * E) / \lambda_{LT}^2$		= 911.511 MPa		
Limiting equivalent slenderness	λ_{L0} = 0.4 * SQRT(π^2 * E / py)		= 30.1976		
Perry coefficient	η_{LT} = (λ_{LT} - λ_{L0}) * 0.007		= 0.11841		
	ϕ_{LT} = [py + (η_{LT} + 1) * pE] / 2		= 687.222 MPa		
Bending buckling strength	pb = pE * py / [ϕ_{LT} + SQRT(ϕ_{LT}^2 - pE * py)]		= 301.621 MPa	< py (355 MPa)	
Buckling resistance moment	Mb = pb * Zx = 436 kNm	>= m_{LT} *Mx = 69.7 kNm		**OK**	16%

TENSION + MOMENT CHECK

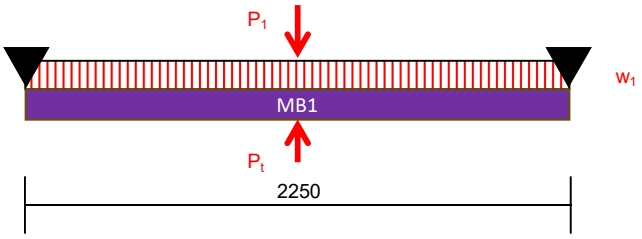
$$\frac{Nt}{Pt} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{69.7}{513} + \frac{0}{170} = 0.14 <= 1$$ **OK**

COMPRESSION + MOMENT CHECK

$$\frac{Nc}{A * py} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{69.7}{513} + \frac{0}{170} = 0.14 <= 1$$ **OK**

LaBTECH		Title: **Checking of Plate Load Test Setup (Max. 750kPa)**	
Date : Oct-2026	Initial : Checked :	Revision : v20261021	Page :

Check: MB1



UC 305x305x97kg/m
S355

Loading

P_t = Test Load
= 1.4 x 67.5
= 94.50 kN

P_1 = Self-weight of concrete block on top supported by middle SB1
= 1.4 x 24 x 0.75 x 2.25 x 1.125
= 63.79 kN (factored)

w_1 = Self-weight of MB1
= 1.4 x 97 x 9.81/1000
= 1.33 kN/m (factored)

Design Shear, V = P/2 + wL/2
= (94.5-63.79) / 2 - 1.33 x 2.25 / 2
= 13.86 kN

Design Moment, M = $PL/4 + wL^2/8$
= (94.5-63.79) x 2.25 / 4 - 1.33 x 2.25^2 / 8
= 16.43 kNm

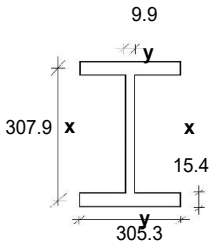
Structural Checking

(STRUCTURAL CHECKING REFER NEXT PAGES)

Calculation Sheet	Job No.	Sheet No.	Rev.
			P.
	Member / Location:	MB1	
Job Tilte :	Drg. Ref.:		
	Made By:	Date: Oct-26	Check By:

DESIGN CODE

Code of Practice for the Structural Use of Steel 2011



MEMBER SPECIFICATION

Member

Steel grade	=	BS S355	
Section	=	UC 305x305x97	(H-section)
Type	=	hot-rolled steel section	

MATERIAL/SECTIONAL PROPERTIES

Modulus of elasticity	E	=	205000	MPa									
Mass	m	=	96.9	kg/m									
Area	A	=	123	cm^2									
Dimensions	D	=	307.9	mm	d	=	246.7	mm					
	B	=	305.3	mm	b	=	152.65	mm					
	T	=	15.4	mm	t	=	9.9	mm					
Moment of inertia	Ix	=	22250	cm^4	Iy	=	7308	cm^4	Iu	= N.A.	cm^4	Iv	= N.A. cm^4
Radius of gyration	rx	=	13.4	cm	ry	=	7.69	cm	ru	= N.A.	cm	rv	= N.A. cm
Elastic modulus	Zx	=	1445	cm^3	Zy	=	479	cm^3					
Plastic modulus	Sx	=	1592	cm^3	Sy	=	726	cm^3					
	Svx	=	150.6	cm^3	Svy	=	717.7						
Torsional modulus	J	=	91.2	cm^4									
Design strength	py	=	355	MPa									

SECTION CLASSIFICATION

Parameter	ε	=	SQRT(275 / py)			=	0.8801	
Stress ratio	r1	=	Nc / (d * t * py)			=	0	
	r2	=	Nc / (A * py)			=	0	
Dimension ratio	b/T	=	9.9	<=	$\frac{15\,\varepsilon}{}$	=	13.2	Flange: Class 3
	d/t	=	24.9	<=	$\frac{80\,\varepsilon}{1+r1}$	=	70.4	Web: Class 1
								Overall: Class 3

DESIGN FORCES AND MOMENTS

Nc (kN)	Nt (kN)	Vx (kN) [along]	Vy (kN) [along]	Mx (kNm) [abt]	My (kNm) [abt]	Tr (kNm)
0.00	0.00	0.00	13.86	16.43	0.00	0.00

SHEAR CAPACITY CHECK

Shear area	Avy	= t * D	=	3048	mm^2					*(Cl. 8.2.1)*	
Shear capacity	Vcy	= py * Avy / SQRT(3)	=	624.76	kN	>=	Vy =	13.86	kN	OK	2%
Shear load check	Vy	<= 0.6 * Vcy	=	374.86	kN		(Low shear case)				
Shear area	Avx	= 0.9 * (2 * B * T)	=	8463	mm^2						
Shear capacity	Vcx	= py * Avx / SQRT(3)	=	1734.6	kN	>=	Vx =	0	kN	OK	0%
Shear load check	Vx	<= 0.6 * Vcx	=	1040.7	kN		(Low shear case)				

MOMENT CAPACITY CHECK

(Cl. 8.2.2)

(Low Shear Case)

Moment capacity	Mcx	= py * Zx	=	512.98	kNm	>=	Mx =	16.43	kNm	OK	3%
Moment capacity	Mcy	= py * Zy	=	170.05	kNm	>=	My =	0	kNm	OK	0%

BIAXIAL MOMENTS: LOCAL CAPACITY CHECK

$$\frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{16.43}{512.98} + \frac{0}{170.045} = 0.03 \quad <= 1$$

OK 3%

SHEAR STRESS AND TORSIONAL STRESS COMBINE CHECK

$$\frac{Vy}{Vcy} + \frac{Tr * T}{J*py} = \frac{13.86}{624.76} + \frac{0.00}{0.324} = 0.02 \quad <= 1$$

OK 2%

Calculation Sheet	Job No.	Sheet No.	Rev.
			P.
	Member / Location:	MB1	
Job Tilte :	Drg. Ref.:		
	Made By:	Date: Oct-26	Check By:

TENSION CAPACITY CHECK

Effective Area $a_e = Ke * a_n$ = 1.1 x(12300 (Ag) - 0 (Area deduction for hole in mm²)) *(Cl. 8.6.1)*

= 13530 mm² >= 12300 mm² (Therefore take Ag)

Connection Type: **Welded Connection**

$a_2 = Ag - a_1$ = 9251.8 mm²

Tension capacity $Pt = py * (a_e - 0 \; a_2)$ = 4366.5 kN >= Nt = 0 kN **OK** 0%

COMPRESSION CAPACITY CHECK *(Cl. 8.7.5)*

Effective length	Lcx	= 2.25 m	Lcy	= 2.25 m	
Slenderness	λx	= Lcx / rx = 16.791	λy	= Lcy / ry = 29.2588	
From Table 8.7, use buckling curve		[b]		[c]	
Robertson constant	α	= 3.5	α	= 5.5	
	pEx	= $(\pi^2 * E) / \lambda x^2$ = 7176	pEy	= $(\pi^2 * E) / \lambda y^2$ = 2363	
Limiting slenderness	λo	= 0.2 * SQRT(π^2 * E / py) = 15.0988			
Perry factor	η	= α * (λx - λ0) / 1000 = 0.006	η	= α * (λy - λ0) / 1000 = 0.078	
	ϕ_C	= [py + (η + 1) * pEx] / 2 = 3787	ϕ_C	= [py + (η + 1) * pEy] / 2 = 1451	
Compressive buckling strength	pcx	= pEx * py / [ϕ_C + SQRT(ϕ_C^2 - pEx * py)] = 352.802 MPa	pcy	= pEy * py / [ϕ_C + SQRT(ϕ_C^2 - pEy * py)] = 325.592 MPa	
Compression resistance	Pc	= A * pcy = 4004.8 kN	>= Nc = 0 kN		**OK** 0%

LATERAL-TORSIONAL BUCKLING CHECK *(Cl. 8.3.1)*

Effective length	L_{LT}	= 2.25 m	m_{LT} = 1		
Slenderness	λ	= L_{LT} / ry	= 29.2588		
Dist. betw. flange shear centers	hs	= D - T	= 292.5 mm		
Web depth	hw	= D - 2 * T	= 277.1 mm		
wraping constant for channel	H	=	= 1.9E+12 mm⁶		
Torsional index	x	= 0.566 * hs * SQRT(A / J)	= 19.2264		
	γ	= 1 - Iy / Ix	= 0.67155		
Buckling parameter	u	= $(4 * Sx^2 * \gamma)/(A^2 * hs^2)]^{0.25}$	= 0.85161		
Slenderness factor	v	= $1 / [1 + 0.05 * (\lambda / x)^2]^{0.25}$	= 0.97298		
	βw	= Zx / Sx	= 0.90766		
Equivalent slenderness	λ_{LT}	= u * v * λ * SQRT(βw)	= 23.0974		
	pE	= $(\pi^2 * E) / \lambda_{LT}^2$	= 3792.51 MPa		
Limiting equivalent slenderness	λ_{L0}	= 0.4 * SQRT(π^2 * E / py)	= 30.1976		
Perry coefficient	η_{LT}	= (λ_{LT} - λ_{L0}) * 0.007	= -0.0497		
	ϕ_{LT}	= [py + (η_{LT} + 1) * pE] / 2	= 1979.51 MPa		
Bending buckling strength	pb	= pE * py / [ϕ_{LT} + SQRT(ϕ_{LT}^2 - pE * py)]	= 375.728 MPa	>= py (355 MPa)	
Buckling resistance moment	Mb	= pb * Zx = 513 kNm	>= m_{LT} *Mx = 16.43 kNm		**OK** 3%

TENSION + MOMENT CHECK

$$\frac{Nt}{Pt} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{16.43}{513} + \frac{0}{170} = 0.03 \;<=\; 1$$ **OK**

COMPRESSION + MOMENT CHECK

$$\frac{Nc}{A * py} + \frac{Mx}{Mcx} + \frac{My}{Mcy} = \frac{0}{4367} + \frac{16.43}{513} + \frac{0}{170} = 0.03 \;<=\; 1$$ **OK**

Web buckling checking (for unstiffened web)

Member **MB1 & SB1**

Section size

Compression force F_x = 94.5 kN

Stiff bearing length b_1 = t + 1.6r + 2T

= 65.02 mm

Section: D = 307.9 mm r = 15.2 mm

B = 305.3 mm

t = 9.9 mm

T = 15.4 mm

d = 246.7 mm p_{yw} = 355 N/mm^2

Check for web-to-flange connection

Connection location: Near ends of member b_e = 0 m

k = 30.6 mm (Rolled section) n = min(2 + 0.6b_e / k , 5)

= 2.00

Bearing capacity P_{bw} of unstiffened web

= (b_1 + nk) t p_{yw}

= 444 kN > F_x **OK**

Check for buckling resistance of web

Loaded flange is restrained against a) rotation relative to web

b) lateral movement relative to other flange: No

Distance from load to end of member a_e = 0 mm < 0.7d

Buckling resistance of unstiffened web P_x

= $25\varepsilon t P_{bw} / \sqrt{(b_1 + nk)\, d}$ x (a_e + 0.7d) / 1.4d

= 274 kN > F_x **OK**

Reduced buckling resistance of unstiffened web P_{xr}

= 0.7d P_x / L_E Effective length of web L_E = 277.1 m

= 171 kN > F_x **OK**

→ Therefore, stiffener in web is not requied.

Appendix B

Standard Loading Schedule

Confirm the actual load and pressure values, loading and unloading sequence, stage durations and observation intervals against the approved project programme before testing.

Plate Load Test	[Sample ID]	
working pressure	350	kPa
300% working pressure	1050.0	kPa
Working load	24.74	kN
300% working load	74.22	kN
Plate diameter	0.30	m
Plate size	0.071	m²
maintain duration	72	hours

	Percentage (%)	Specified Pressure (kPa)	Applied load (kN)	Allowable Criteria
1st cycle	0	0.0	0.00	take readings only
	50	175.0	12.37	less than 0.05mm in 10 minutes
	100	350.0	24.74	
	50	175.0	12.37	take readings only
	0	0.0	0.00	less than 0.05mm in 10 minutes

	Percentage (%)	Specified Pressure (kPa)	Applied load (kN)	Allowable Criteria
2nd cycle	0	0.0	0.00	take readings only
	50	175.0	12.37	less than 0.05mm in 10 minutes
	100	350.0	24.74	
	150	525.0	37.11	
	200	700.0	49.48	
	150	525.0	37.11	take readings only
	100	350.0	24.74	
	50	175.0	12.37	
	0	0.0	0.00	less than 0.05mm in 10 minutes

	Percentage (%)	Specified Pressure (kPa)	Applied load (kN)	Allowable Criteria
3rd cycle	0	0.0	0.00	take readings only
	50	175.0	12.37	less than 0.05mm in 10 minutes
	100	350.0	24.74	
	150	525.0	37.11	
	200	700.0	49.48	
	250	875.0	61.85	
	300	1050.0	74.22	maintain for 72 hours with readings taken at 1000, 1300 and 1700 each day
	250	875.0	61.85	take readings only
	200	700.0	49.48	
	150	525.0	37.11	
	100	350.0	24.74	
	50	175.0	12.37	
	0	0.0	0.00	less than 0.05mm in 10 minutes

Appendix C

Sample Test Record Sheet

LaBTECH

WORKSHEET FOR DETERMINATION OF VERTICAL DEFORMATION AND STRENGTH CHARACTERISTICS OF SOIL BY PLATE LOADING TEST (INCREMENTAL LOADING METHOD)

Test Results (cont'd) :

Plate Load Test No. : _____

Date	Time	Temp. (°C)	Applied Pressure (kPa)	Applied Load (kN)	Load Cell (kN)	Dial Gauge Reading (mm) (0.01 mm per Division)					Cumulative Deformation (mm)
						Dial Gauge No.				Average	
						1	2	3	4		